

Hardy-Weinberg Equilibrium

Hardy-Weinberg law is the fundamental law of population genetics, which provides the basis for studying the Mendelian population. According to this law,

- (i) Mutations introduce new genes into a species resulting into a change in gene frequencies.
- (ii) In certain conditions gene frequencies remain constant. The conditions necessary for gene frequency to remain constant are:
 - Mating must be completely random.

- Mutation must not occur.
- The population must be very large.
- All genes must have an equal chance of being passed to the next generations.

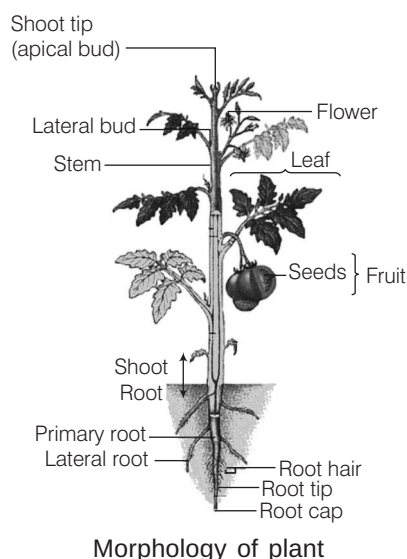
- (iii) Constant gene frequencies over several generations indicate that natural selection and evolution are not taking place.
- (iv) Changing gene frequencies would indicate that evolution is in progress.

PLANT MORPHOLOGY AND PHYSIOLOGY

Angiosperms are the seed bearing, flowering plants in which seeds are formed inside fruits; and male and female reproductive parts are organised into flowers.

MORPHOLOGY OF ANGIOSPERMS

Morphology (*Morphe*—form; *logos*—study) is the branch of biology, which deals with the study of external forms and features of different plant organs like roots, stems, leaves, flowers, seeds, fruits, etc. The body of typical angiospermic plant is differentiated into an underground root system and an aerial shoot system.



Morphology of plant

Types of Plants

On the basis of nutrition, plants are of following types:

- (i) **Autotrophic plants** These are green and capable of making their own food by the process of photosynthesis, e.g. all green plants.
- (ii) **Parasitic plants** Depend on other plants for food and water. They have special roots for absorption of food and water. These roots are called **sucking roots** or **haustoria**, e.g. *Cuscuta*, *Duranta*.

Type of parasitic plants are as follow:

- (i) **Obligate parasite**, e.g. *Nuytsia floribunda*.
- (ii) **Facultative parasite**, e.g. *Rhinanthus*.
- (iii) **Root parasite**, e.g. *Rafflesia*, sandal wood.
- (iv) **Stem parasite**, e.g. *Cuscuta*, *Loranthus*.
- (v) **Saprophytic humus plants** Grow on dead organic matter, e.g. fungi.
- (vi) **Symbiotic plants** Symbiosis or mutually beneficial partnership of two organisms, e.g. mycorrhiza associated with roots of higher plants and legumes associated with *Rhizobium* (N_2 -fixing bacteria).

Different organ systems present in angiosperms are as follow:

Root System

Roots develop from radicle of seed. They are non-green, underground, geotropic, phototropic and hydrotropic in nature and they do not bear buds, nodes and internodes. They have unicellular root hairs. Lateral roots arise endogenously from pericycle.

Types of Roots

Roots are of two types:

- (i) **Tap root** It develops from radicle. The primary root grows and gives rise to secondary and tertiary roots forming tap root system, e.g. dicot.
- (ii) **Adventitious root** These roots develop from any part of the plant body other than the radicle, e.g. monocot.

Functions of Roots

- Roots help in fixation and absorption of water and minerals.
- Storage of food is one of the major function of roots, conduction of water takes place by roots.
- It helps in photosynthesis and respiration by providing supply of water to all parts of plant.

Shoot System

Shoot system consists of following parts:

Stem

Stem is the ascending part, of plant, bearing branches, leaves, flowers and fruits, formed by the prolongation of the plumule of embryo. It bears nodes and internodes and has buds. It may also bear multicellular hairs on external surface.

Modifications of Stem

Stem is modified in the following forms:

- **Rhizome** It grows parallel or horizontal to soil surface. It bears nodes, internodes, buds and leaves, e.g. ginger.
- **Tuber** It is the terminal portion of underground stem branch, which is swollen due to accumulation of food, e.g. potato.
- **Corm** It grows vertically to soil surface and is covered by thin sheathing leaf bases of dead leaves called **scales**, e.g. *Colocasia*, *Gladiolus*.
- **Phylloclade** It is green, flattened or rounded succulent stem with leaves either feebly developed or modified into spines, e.g. *Opuntia*, *Euphorbia*.
- **Cladode** Phylloclade with one or two internodes are called cladode, e.g. *Asparagus*.
- **Thorn** It is the modification of axillary bud. Thorns not only reduce transpiration, but also check browsing by animals, e.g. *Citrus*, *Duranta*.

Leaves

A leaf is a flat, lateral outgrowth of the stem or the branch, arising from a node and usually having a bud in its axil. It originates from shoot apical meristem. These are the most important organ for photosynthesis. The chief functions of the leaf include photosynthesis and transpiration.

Venation

The arrangement of veins and the veinlets in the lamina of leaf is called venation. These are of two main types:

- **Reticulate venation** The branches or veins form a network, e.g. dicots.
- **Parallel venation** The veins and veinlets remain parallel to each other, e.g. monocots.

Phyllotaxy

It is the arrangement of leaves on the stem or branch and is adopted, so that each leaf is properly exposed to sunlight.

Types of Phyllotaxy

Phyllotaxy is usually of three types:

- **Alternate** When a single leaf arises at each node it is called as alternate, e.g. *Hibiscus* sp.
- **Opposite** When two leaves arise at each node and occurs opposite to each other it is called as opposite, e.g. *Syzygium*, *Calotropis*.
- **Whorled** If more than two leaves arise at a node and form a whorl. Heterophylly is occurrence of more than one type of leaves on the same plant, e.g. *Ranunculus* sp.

Flower

A flower is the reproductive unit in angiosperms, which is meant for sexual reproduction. A flower arises in the axil of a leaf-like structure called **bract**, e.g. *Bougainvillea*.

Structure of a Flower

Flower consists of four distinct parts, calyx, corolla, androecium and gynoecium.

- The calyx and corolla are accessory parts while the androecium and gynoecium are essential parts.
- When calyx, corolla, androecium and gynoecium are all present on same flower it is called as **complete flower**.
- A flower can be **unisexual** or **bisexual**.

- The symmetry of a flower depends upon the size, shape and arrangement of floral parts.
- On the basis of symmetry flower can be:

- (i) **Actinomorphic** When a flower can be divided into two exactly equal halves by any verticle section passing through the centre of a flower, e.g. mustard.
- (ii) **Zygomorphic** When a flower can be divided into two identical halves through only one particular plane, e.g. *Cassia*.

- An extra whorl of sepal-like green coloured structures called **bracteoles** occurs on the outside of calyx in members of Malvaceae. It is called as **epicalyx** fused sepals and petals form **perianth**.
- Stamens and pistils are the male and female reproductive organs of a flower. Stamen consists of anthers and filaments while pistil consists of stigma, style and ovary.
- Anther consisting of four microsporangia (tetrasporangiate) is called **dithecal**. Anthers are reniform or kidney-shaped and consisting of two microsporangia (bisporangiate) is called as **monotheical**.
- In Cruciferae, stamens are six (2 outer shorts and 4 inner longs). This condition is called **tetradynamous**.
- **Monoadelphous stamens** are found in members of Malvaceae.
- **Axile placentation** is found in members of Malvaceae, Liliaceae and Solanaceae.
- **Basal placentation** is present in members of Asteraceae.

Types of Flower

On the basis of position of ovary in flowers, these can be of different types such as:

- (i) When the ovary is at the top of thalamus, flowers are **hypogynous** and ovary is superior, e.g. *Brassica*, *Citrus*.

- (ii) When calyx and corolla arise from the upper side of ovary and surrounded by thalamus, ovary is inferior and flower is **epigynous**, e.g. *Cucurbita*, Sunflower.
- (iii) When ovary is half superior and half inferior then it is called as **perigynous**, e.g. rose, plum, etc.

The flowers, in which both male and female reproductive organs are present and called as bisexual flowers, whereas the flowers having either male or female reproductive organ are called as unisexual flowers.

On the basis of the presence of male and female flowers, plants are of following types:

- (i) The plants, in which both male and female flowers are present are called as **monoecious plants**, e.g. *Cocos*, *Ricinus*, *Zea*, *Colocasia*, *Acalypha*.
- (ii) The plants, in which either male or female flowers are found, are called as **dioecious plants**, e.g. mulberry, papaya.
- (iii) When unisexual (male or female), bisexual flowers are present on the same plant the plant is called **polygamous plants**, e.g. *Polygonum*, mango.

Largest flower is *Rafflesia* and smallest is *Wolffia*.

Inflorescence

The arrangement of flowers and mode of distribution of flowers on the shoot system of a plant is called **inflorescence**.

Types of Inflorescence

- (i) **Racemose** (Indefinite) Main axis of inflorescence does not end in a flower, but continues to grow. The development of flowers is acropetal. The opening of flowers is centripetal.
- (ii) **Cymose** (Definite) Main axis ends in a flower. The development of flowers is basipetal and opening of flowers is centrifugal.
- (iii) **Special type capitulum** It is characteristic type of inflorescence in members of Compositae (Asteraceae).

Fruits

A fruit is a part of a flowering plant that is derived from specific tissues of the flower. The ovary of the flower develops into the fruit. It is the characteristic feature of flowering plant. The term 'fruit' refers to the mature or ripened ovary that contains seeds.

Parthenocarp is the formation of fruits without fertilisation, such fruits are **seedless**.

Types of Fruits

Fruits can be broadly classified into following types:

- (i) **Simple fruits** These are the fruits develop from the single simple or compound ovary of a flower. These can be dry (pericarp dry) or fleshy fruits.
- (ii) **Aggregate fruits** They are the group of fruitlets, which develops from a flower having polycarpellary free gynoecium, also called **etaerio**.
- (iii) **Multiple (Composite) fruits** These are composite fruits, which develop from an entire inflorescence.

Types and Edible Parts of some Common Fruits

Common name	Botanical name	Type	Edible parts
Simple Fruits			
Mango	<i>Mangifera indica</i>	Drupe	Fleshy mesocarp
Coconut	<i>Cocos nucifera</i>	Drupe	Endosperm
Almond	<i>Prunus amygdalus</i>	Drupe	Seeds
Walnut	<i>Juglans regia</i>	Drupe	Cotyledons
Apple	<i>Pyrus malus</i>	Pome	Fleshy thalamus
Pear	<i>Pyrus communis</i>	Pome	Fleshy thalamus
Betel nut	<i>Areca catechu</i>	Berry	Seeds
Pea	<i>Pisum sativum</i>	Legume	Seeds
Cashew nut	<i>Anacardium occidentale</i>	Nut	Cotyledons and fleshy thalamus
Litchi	<i>Litchi chinensis</i>	Nut	Aril
Water chestnut	<i>Trapa bispinosa</i>	Nut	Seeds
Pomegranate	<i>Punica granatum</i>	Balausta	Succulent testa
Bengal quince (bael)	<i>Aegle marmelos</i>	Amphisarca	Inner fleshy layer of pericarp and placenta
Grape	<i>Vitis vinifera</i>	Berry	Pericarp and placenta
Papaya	<i>Carica papaya</i>	Berry	Mesocarp
Tomato	<i>Solanum lycopersicum</i>	Berry	Mesocarp
Banana	<i>Musa paradisiaca</i> sp.	Berry	Pericarp and placenta
Watermelon	<i>Citrullus lanatus</i>	Pepo	Mesocarp and endocarp
Lemon	<i>Citrus limon</i>	Hesperidium	Juicy placenta and juicy hairs developed from endocarp
Wheat	<i>Triticum aestivum</i>	Caryopsis	Starchy endosperm
Aggregate Fruits			
Lotus	<i>Nelumbo nucifera</i>	Etaerio of achenes	Fleshy thalamus and seeds
Strawberry	<i>Fragaria vesca</i>	Etaerio of achenes	Fleshy thalamus and seeds
Custard apple	<i>Anona squamosa</i>	Etaerio of berries	Inner layer of pericarp and thalamus
Multiple or Composite Fruits			
Mulberry	<i>Morus alba</i> and <i>M. nigra</i>	Sorosis	Succulent perianth and fleshy axis
Pineapple	<i>Ananas comosus</i>	Sorosis	Fleshy axis, bracts, fused peria and pericarp
Jack fruit	<i>Artocarpus heterophyllus</i>	Sorosis	Fleshy axis, bracts, perianth and seeds
Fig	<i>Ficus carica</i>	Syconus	Fleshy receptacle or thalamus

Seeds

A seed is an embryonic plant enclosed in a protective outer covering called **seed coat**, usually with some stored food. The ovary of the flower contains the ovules. The fertilised ovules develop into seeds. The study of seed is called **Spermology**. The seeds are of mainly two types:

- (i) Non-endospermic or exalbuminous.
- (ii) Endospermic or albuminous.
- **Vivipary** When the seeds of a plant germinate before they detach from the parent, e.g. mangroves.
- **Parthenogenesis** When an unfertilised egg develops into embryo (false embryo), the process is called **parthenogenesis**.
- **Parthenogamy** When sexual reproduction fails, the gametes directly behave as spores, these are called **parthenospores** or **zygospores**. The **phenomenon** is called parthenogamy.
- **Polyembryony** The occurrence of more than one embryo in the seed is called **polyembryony**.

Plant Anatomy

The branch of botany dealing with the internal organisation of plants is called **anatomy**. **N Grew** (1682), 'Father of Plant Anatomy', gave the terms tissue and parenchyma. **Nageli** (1850) coined the terms meristem, xylem and phloem. A group of similar or dissimilar cells that perform a common function and have a common origin is called **tissue**. The tissues have been classified into two groups:

- (i) **Meristematic tissues** constitute an undifferentiated mass of cells, that are in a continuous state of division or retain their power of division.
 - These tissues divide to form new cells, which differentiate to give rise to permanent tissues.
 - Procambium is the meristematic tissue, which forms primary xylem, phloem and vascular cambium.
- (ii) **Permanent tissues** do not have the power of division. They include protective tissue (e.g. epidermis), supporting tissue (e.g. parenchyma, collenchyma, sclerenchyma) and conducting tissue (e.g. xylem, phloem).
- Phloem, consists of three types of cells are called **sieve element**, companion cells and phloem parenchyma.
 - Xylem consists of tracheids, vessels, xylem parenchyma and xylem fibres.
- **Note** *Marchantia* belongs to Bryophyta and conducting tissues, i.e. xylem and phloem are absent in them for water absorption.
- In monocot leaf, mesophyll is undifferentiated and made up of only spongy parenchyma.
- In bicollateral vascular bundles, xylem is sandwiched between external and internal phloem.

- Parenchyma is a simple permanent tissue. When cells of this tissue have large intercellular spaces they are called as **aerenchyma**, e.g. hydrophytes.

PLANT PHYSIOLOGY

Plant physiology is concerned with the functioning of plants. It is the science, which is connected to the material and energy exchange, growth and development as well as movement of plant.

Plant Nutrients

The minerals required for proper growth and development plants are called as **nutrients**. There are two types of nutrients:

- (i) **Macronutrients** Minerals needed in large amount, e.g. C, H, O, Ca, P, Mg, S, K, N.
- (ii) **Micronutrients** Minerals needed by plants in traces, e.g. Fe, Zn, Cl, Cu, Mo, B, Mn, Ni.
- C, O, H and N are non-mineral elements that are abundant in plants.

Imbibition

It is the phenomenon of absorption of water or liquid by the solid particles of a substances without forming a solution.

- The solid particles, which absorb water are called **imbibants**. These are hydrophilic colloids.
- The liquid, which is imbibed is called **imbibate**. The pressure that an imbibant develops after being imbibed is called **imbibition pressure/matric potential** (Ψ_m).
- Increase in the pressure will increase the imbibition.

Diffusion

It is the net movement of particles from a region of its higher concentration to a region of its lower concentration due to their own kinetic energy.

- Rate of diffusion $\propto$ temperature $\propto \frac{1}{\text{Size or mass of particle}}$
- **Diffusion Pressure Deficit (DPD)** The difference between diffusion pressure of the solution and its solvent at a particular temperature and atmospheric conditions, is called DPD.

Permeability

Permeability is the ability of a membrane to permits the movement of molecules across it.

Types of membrane on the basis of permeability are:

1. **Permeable membrane** Allows the diffusion of both the solvent and solute molecules across it.
2. **Impermeable membrane** Neither the solute nor solvent molecules can diffuse across membrane.

3. **Semipermeable membrane** Allows movement of solvent molecules across it, but not the solute molecules.
4. **Selectively permeable membrane** Membrane allowing diffusion of only selected molecules across it.

Osmosis

It is the movement of solvent (water) molecules from a region of lower concentration to a region of higher concentration across a semipermeable membrane. It was discovered by **Nollet** in 1748.

Types of Osmosis

Osmosis are of following types:

- (i) When a cell is placed in hypotonic (less concentrated) solution water enters into the cell. This is called as **endosmosis**.
 - (ii) When a cell is placed in hypertonic (high concentrated) solution water exits the cell. It is called as **exosmosis**.
- Isotonic solutions are the solutions of same concentration. No net movement of molecules occurs between isotonic solution.
 - The actual pressure that develops in a solution, when it is separated from pure water by means of a semipermeable membrane is called **osmotic pressure**.
 - Osmotic pressure of a plant cell can vary from 4-5 atm.

Turgor Pressure

As a result of osmosis, water enters into the cell sap. Due to this a pressure is developed by cell sap against cell wall making the cell turgid. This pressure is called **turgor pressure/hydrostatic pressure/pressure gradient (ψ_p)**.

Where, $\psi_p = \psi_s$
 ψ_s = Concentration gradient.

Plasmolysis and Deplasmolysis

The phenomenon of shrinkage of protoplasm away from the cell wall under the influence of hypertonic solution is called **plasmolysis**. **Deplasmolysis** occurs enters into the cell sap, due to this the cell becomes turgid resulting into protoplasm that assumes its normal shape.

TRANSPIRATION

The loss of water vapour from the living tissues of aerial parts of plant into the atmosphere is referred to as **transpiration**.

- The transpiration occurs from the leaves all the time and water released from leaves absorbs heat from the surrounding atmosphere to vapourise (similar to evaporation) due to which, area becomes cool nearby.
- It is also the ultimate cause of water movement in a plant stem against the gravity.

Types of Transpiration

Transpiration are of following types:

- (i) **Stomatal transpiration** Stomata is a tiny pore that is used for gas exchange. Air containing CO_2 and O_2 enters the plant through these openings and utilised by the plant. It accounts for 80-90% of total water loss from the plant.
- (ii) **Cuticular transpiration** Cuticle is the relatively impermeable covering of plant. If it is thin and green then upto 10-20% transpiration take place through it.
- (iii) **Lenticular transpiration** Transpiration through lenticels, i.e. an airy aggregation of cells within the structural surfaces of stem, roots and other parts of vascular plants. Very little transpiration occurs through lenticels.

Factors Affecting Rate of Transpiration

Two factors affect rate of transpiration, which are as follow:

External Factors

- **Temperature** Transpiration increases with increase in atmospheric temperature.
- **Light** Rate of transpiration, i.e. rate of transpiration increases in light and decreases in dark.
- **Wind velocity** Upto 20-30 km/hr the rate of transpiration increases, while wind velocity of 40-50 km/hr decreases transpiration.
- **Humidity** The rate of transpiration is inversely proportional to the relative humidity.

$$\text{Transpiration} \propto \frac{1}{\text{Humidity}}$$

Internal Factors

- **Area of leaf** Rate of transpiration is directly proportional to the area of leaf. $\text{Transpiration} \propto \text{Area of leaf}$
- **Leaf structure** Some modifications of leaf structure such as thick cuticle on surface of plant parts, wax, resin and suberin coating on plants, sunken stomata in leaves contributes to reduction in transpiration.

PHOTOSYNTHESIS

Photosynthesis (Gk. *photon*–light; *synthesis*–putting together) is the anabolic process by which green plants synthesise complex carbohydrates from simple substances like carbon dioxide and water with the help of light energy.

